

Practice

Paper 21

A කොටස

$$\boxed{01} \quad \sum_{r=1}^n (r-1)(r+2) = \frac{n(n-1)(n+4)}{3}$$

* $n=1$ වීද ඉක්මුලය

$$\begin{aligned} \text{L.H.S.} &= \sum_{r=1}^1 (r-1)(r+2) \\ &= (1-1)(1+2) = 0 \end{aligned}$$

$$\begin{aligned} \text{R.H.S.} &= \frac{n(n-1)(n+4)}{3} \\ &= \frac{1(1-1)(1+4)}{3} = 0 = \text{L.H.S.} \quad (05) \end{aligned}$$

* $n=p$ වීද ඉක්මුලය ඉතර බව දරනලේඡනය කරමු.

$$\sum_{r=1}^p (r-1)(r+2) = \frac{p(p-1)(p+4)}{3} \quad (05)$$

* $n=p+1$ වීද ඉක්මුලය

$$\begin{aligned} \text{L.H.S.} &= \sum_{r=1}^{p+1} (r-1)(r+2) \\ &= \sum_{r=1}^p (r-1)(r+2) + [(p+1-1)(p+1+2)] \quad (05) \\ &= \frac{p(p-1)(p+4)}{3} + p(p+3) \\ &= \frac{p}{3} [(p-1)(p+4) + 3(p+3)] \end{aligned}$$

$$\begin{aligned}
 \text{L.H.S.} &= \frac{p}{3} [p^2 + 3p - 4 + 3p + 9] \\
 &= \frac{p}{3} [p^2 + 6p + 5] \\
 &= \frac{p}{3} [(p+5)(p+1)] \\
 &= \frac{(p+1)[(p+1)-1][p+1+4]}{3} \quad (05)
 \end{aligned}$$

$$= \text{R.H.S.}$$

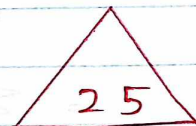
* $n = 1$ විට ත්‍රිකෝණයේ ඡේදය බව පෙන්වා දීම.

* $n = p$ විට ත්‍රිකෝණයේ ඡේදය බව පෙන්වා දීම.

* $n = p+1$ විට ත්‍රිකෝණයේ ඡේදය බව පෙන්වා දීම.

$$\therefore \forall n \in \mathbb{Z}^+ \text{ සඳහා } \sum_{r=1}^n (r-1)(r+2) = \frac{n(n-1)(n+4)}{3}$$

මු. (05)



02

$$|3x - 5| \geq 2x + 3$$

$$\rightarrow 3x - 5 = 0$$

$$x = 5/3$$

$$* \quad x \leq 5/3$$

$$-(3x - 5) \geq 2x + 3$$

$$-3x + 5 \geq 2x + 3$$

$$5x \leq 2$$

$$x \leq \frac{2}{5}$$

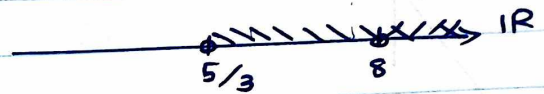


$$x \in (-\infty, 2/5] \rightarrow \textcircled{1}$$

$$* \quad x \geq 5/3$$

$$(3x - 5) \geq 2x + 3$$

$$x \geq 8$$



$$x \in [8, \infty) \rightarrow \textcircled{2}$$

$$\text{விடை} = x \in (-\infty, 2/5] \cup [8, +\infty)$$

$$\text{OR} \quad \{x / x \leq 2/5 \text{ or } x \geq 8\} \quad (05)$$

$$x \text{ எதுவும் } -x \text{ எதுவும்} \quad (05)$$

$$|-3x - 5| \geq -2x + 3$$

$$|3x + 5| \geq 3 - 2x$$

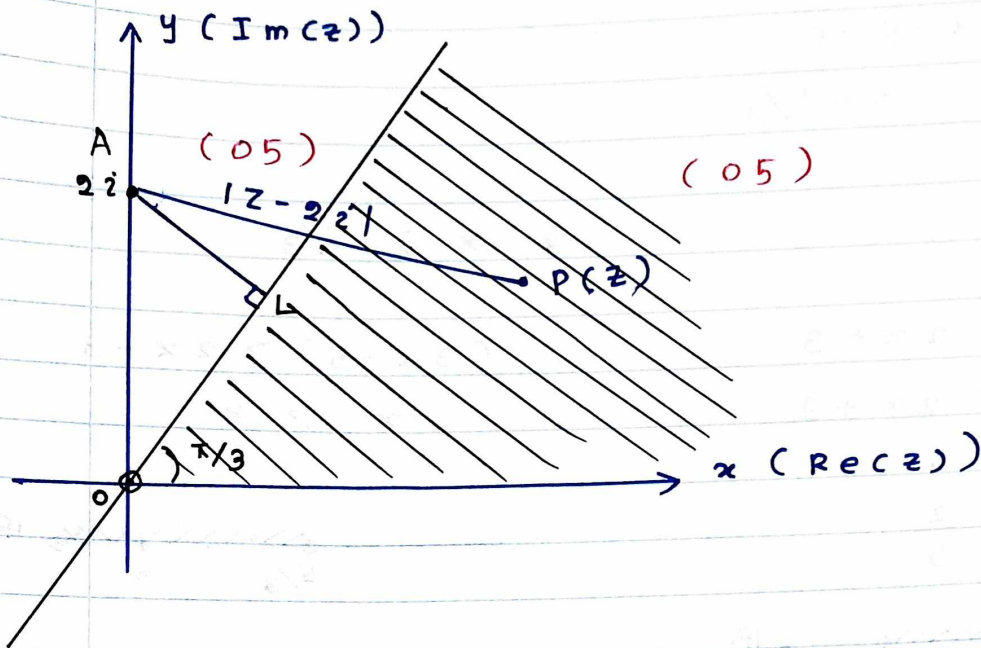
$$\Downarrow$$

$$\{x / x \geq -2/5 \text{ or } x \leq -8\} \quad (05)$$

25

03

$$\therefore 0 \leq \operatorname{Arg} z \leq \pi/3$$



$$|z - 2i|$$

$$\rightarrow A = 2i$$

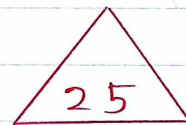
$$|z - 2i| = AP \quad (05)$$

$$|z - 2i|_{\min} = AL \quad (05)$$

$$= 2 \sin \pi/6$$

$$= 2 \times \frac{1}{2}$$

$$= \underline{\underline{1}} \quad (05)$$



04

$$\left(x^3 - \frac{1}{x^2}\right)^n = \sum_{r=0}^n {}^nC_r (x^3)^{n-r} \left(\frac{-1}{x^2}\right)^r$$

$$= \sum_{r=0}^n {}^nC_r (-1)^r x^{3n-5r} \quad (05)$$

 x^5 ಇದ್ದಾಗ, r

$$3n - 5r = 5$$

$$r = \frac{3n-5}{5} = p \rightarrow (1)$$

 x^{10} ಇದ್ದಾಗ, r

$$3n - 5r = 10$$

$$r = \frac{3n-10}{5} = q \rightarrow (2)$$

$$p - q = 1 \quad (05)$$

$${}^nC_p (-1)^p + {}^nC_q (-1)^q = 0 \quad (05)$$

$${}^nC_p (-1)^p + {}^nC_{p-1} (-1)^{p-1} = 0$$

$$\frac{n!}{p!(n-p)!} (-1)^p - \frac{n!}{(p-1)!(n-p+1)!} (-1)^{p-1} = 0$$

$$p = n - p + 1$$

$$p = \frac{n+1}{2} \rightarrow (3)$$

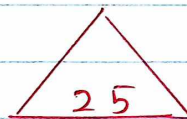
$$(1) = (3)$$

$$\frac{n+1}{2} = \frac{3n-5}{5} \quad (05)$$

$$5n+5 = 6n-10$$

$$15 = n$$

$$\underline{n = 15} \quad (05)$$



05

$$\lim_{x \rightarrow 0} \frac{1 - \cos\left(\frac{k\pi x}{4}\right)}{x^2(x+9)} = \frac{\pi^2}{32}$$

$$\lim_{x \rightarrow 0} \frac{2 \sin^2\left(\frac{k\pi x}{8}\right)}{x^2} \times \lim_{x \rightarrow 0} \frac{1}{x+9} = \frac{\pi^2}{32} \quad (05)$$

$$2 \times \left[\lim_{x \rightarrow 0} \frac{\sin\left(\frac{k\pi x}{8}\right)}{\left(\frac{k\pi x}{8}\right)} \right]^2 \times \frac{k^2 \pi^2}{64} \times \frac{1}{9} = \frac{\pi^2}{32} \quad (05)$$

$$2 \times (1)^2 \times \frac{k^2}{2} \times \frac{1}{9} = 1 \quad (05)$$

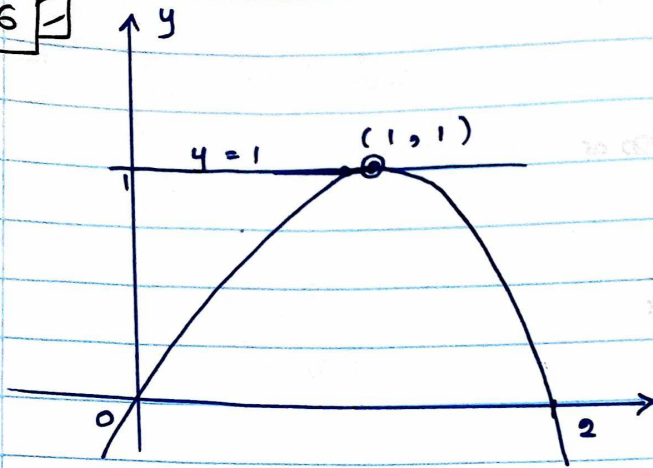
$$k^2 = 9$$

$$k = \pm 3$$

$$k > 0 \text{ then } \underline{k = 3} \quad (05)$$

25

06



$$V = \pi (1)^2 \cdot 2 - 2 \int_0^1 \pi (1-y)^2 dx \quad (05)$$

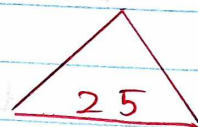
$$1-y = x^2 - 2x + 1 = (x-1)^2$$

$$V = 2\pi - 2\pi \int_0^1 (x-1)^4 dx \quad (05)$$

$$V = 2\pi - 2\pi \left[\frac{(x-1)^5}{5} \right]_0^1 \quad (05)$$

$$V = 2\pi - 2\pi \left(\frac{1}{5} - 0 \right) \quad (05)$$

$$V = \underline{\underline{\frac{8\pi}{5}}}$$



07

$$\frac{x^2}{3} - \frac{y^2}{2} = 1$$

x විෂයෙන් අනුකලනය

$$\frac{2x}{3} = \frac{2y}{2} \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{3y}$$

$$\left(\frac{dy}{dx} \right)_p = m = \frac{2\sqrt{3} \sec \theta}{3\sqrt{2} \tan \theta} \quad (05)$$

විෂර්ෂයේ සමීකරණය

$$\frac{y - \sqrt{2} \tan \theta}{x - \sqrt{3} \sec \theta} = \frac{2\sqrt{3} \sec \theta}{3\sqrt{2} \tan \theta}$$

$$3\sqrt{2} y \tan \theta - 6 \tan^2 \theta = 2\sqrt{3} x \sec \theta - 6 \sec^2 \theta \quad (05)$$

$$2\sqrt{3} x \sec \theta - 3\sqrt{2} y \tan \theta = 6$$

$$\frac{x \sec \theta}{\sqrt{3}} - \frac{y \tan \theta}{\sqrt{2}} = 1$$

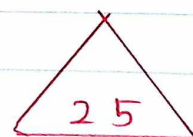
$$\text{අනුකලනයේ අනුපාතය} = - \frac{3\sqrt{2} \tan \theta}{2\sqrt{3} \sec \theta}$$

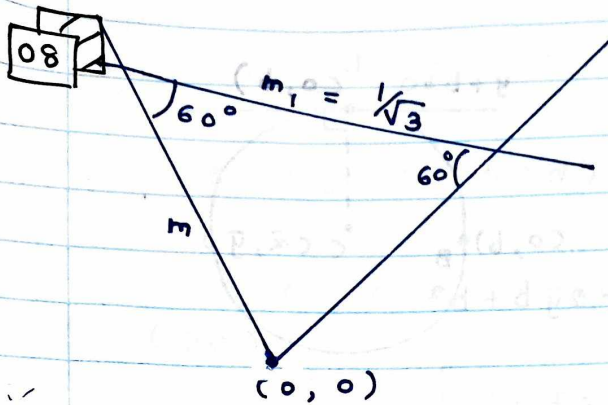
(05)

$$\frac{5/2 - \sqrt{2} \tan \theta}{0 - \sqrt{3} \sec \theta} = \frac{-3\sqrt{2} \tan \theta}{2\sqrt{3} \sec \theta}$$

$$\tan \theta = \frac{1}{\sqrt{2}} \quad (05)$$

$$(\because 0 < \theta < \pi/2) \quad \theta = \pi/4 \quad (05)$$





$$\tan \theta = \left| \frac{m_2 - m_1}{1 + m_1 m_2} \right|$$

$$\tan 60^\circ = \left| \frac{m - \frac{1}{\sqrt{3}}}{1 + m \frac{1}{\sqrt{3}}} \right| \quad (05)$$

$$\sqrt{3} = \frac{\pm (m - \frac{1}{\sqrt{3}})}{(1 + m/\sqrt{3})} \quad (05)$$

$$(+)\quad \sqrt{3} + m = m - \frac{1}{\sqrt{3}}$$

(05)

m ඉඳහා අගයන් නොලැබේ.

$$(-)\quad \sqrt{3} + m = -m - \frac{1}{\sqrt{3}}$$

$$2m = -\frac{2}{\sqrt{3}}$$

$$m = -\frac{1}{\sqrt{3}}$$

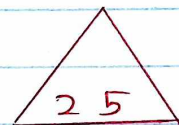
(05)

$\therefore$ රේඛාවේ සමීකරණ

$$\underline{\underline{x = 0}}$$

$$\underline{\underline{y = -\frac{1}{\sqrt{3}} x}}$$

(05)





$$CB = CL \quad (05)$$

$$(\bar{x} - 0)^2 + (\bar{y} - b)^2 = (\bar{y} + b)^2 \quad (05)$$

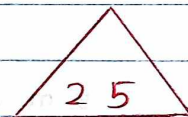
$$\bar{x}^2 + \bar{y}^2 - 2\bar{y}b + b^2 = \bar{y}^2 + 2\bar{y}b + b^2$$

$$\bar{x}^2 = 4\bar{y}b \quad (05)$$

$\bar{x}$ හා $\bar{y}$ භූමිකා x, y ලෙස නැවත නම් කළ විට

(05)

$$\underline{\underline{x^2 = 4by}}$$





$$\cos 6\theta + \cos 4\theta + \cos 2\theta + 1 = 0$$

$$2\cos 5\theta \cos \theta + 2\cos^2 \theta - 1 + 1 = 0 \quad (05)$$

$$(05)$$

$$\cos \theta (\cos 5\theta + \cos \theta) = 0$$

$$\cos \theta [2\cos 3\theta \cos 2\theta] = 0 \quad (05)$$

$$\cos \theta \cos 2\theta \cos 3\theta = 0$$

$$\cos \theta = 0 \quad \cos 2\theta = 0 \quad \cos 3\theta = 0$$

$$\cos \theta = \cos \pi/2 \quad \cos 2\theta = \cos \pi/2 \quad \cos 3\theta = \cos \pi/2$$

$$(05)$$

$$\theta = 2h\pi \pm \pi/2$$

$$(h \in \mathbb{Z})$$

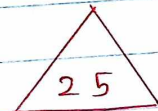
$$\theta = n'\pi \pm \frac{\pi}{4}$$

$$(n' \in \mathbb{Z})$$

$$\theta = \frac{2m\pi}{3} \pm \frac{\pi}{6}$$

$$(m \in \mathbb{Z})$$

(05)



B කොටස



a) $px^2 + qx + r = 0$ මූල තාත්වික යුග්මයක් වන බව ඉක්මනින් පෙන්වන්න (α, β)

$$\Delta > 0 \quad \alpha + \beta = -\frac{q}{p} > 0 \rightarrow (2) \quad (05)$$

$$q^2 - 4pr > 0 \rightarrow (1) \quad \alpha\beta = \frac{r}{p} > 0 \rightarrow (3)$$

$$p^2x^2 + p(3q-2r)x + (2q-r)(q-r) + pr = 0 \quad (\tau, \delta)$$

$$\Delta = B^2 - 4AC$$

$$= [p(3q-2r)]^2 - 4p^2[(2q-r)(q-r) + pr] \quad (05)$$

$$= p^2(3q-2r)^2 - 4p^2[2q^2 - 3qr + pr + r^2]$$

$$= p^2[9q^2 - 12qr + 4r^2 - 4(2q^2 - 2qr + r^2)]$$

$$= p^2[9q^2 - 12qr + 4r^2 - 8q^2 + 8qr - 4r^2] \quad (05)$$

$$= p^2[q^2 - 4qr] \quad (05)$$

$$= \underbrace{p^2}_{(+)} \underbrace{[q^2 - 4qr]}_{(+) \text{ ① න් } \downarrow} \quad (05)$$

$$\Delta \Rightarrow (+) \text{ වේ. } (05) \quad (\because p \neq 0)$$

$\therefore$ මූල තාත්වික යුග්මයක් වේ. නැති නම් මූල දෙක ඉඳිවේ.

$$*\tau + \delta = \frac{p(2r-3q)}{p^2} \quad (05)$$

$$= \frac{2r}{p} - \frac{3q}{p} \quad (05)$$

$$\text{② න් } \text{③ න් } \downarrow \quad (+) + (+) \Rightarrow (+)$$

$$*\tau\delta = \frac{(2q^2 - 2qr + r^2)}{p^2} \quad (05)$$

$$= \frac{2q^2}{p^2} - \frac{2qr}{p^2} + \frac{r^2}{p^2} \quad (10)$$

$$\text{② න් } \text{③ න් } \text{③ න් } = (+) + (+) + (+) \Rightarrow (+)$$

b) $f(x) = x^2 + 2xy + 3y^2 - 6x - 2y + 11$

$$= x^2 + 2x(y-3) + (3y^2 - 2y + 11) \quad (05)$$

මෙය x නි ඉරිතේ x ජිතරනය.

x^2 නි ඉරිතරනය > 0 තේ. x නි
(05)

$$\Delta = B^2 - 4AC$$

$$= [2(y-3)]^2 - 4(1)(3y^2 - 2y + 11) \quad (05)$$

$$= 4[y^2 + 9 - 6y - 3y^2 + 2y - 11]$$

$$= 4[-2y^2 - 4y - 2]$$

$$= -8[y^2 + 2y + 1]$$

$$= -8(y+1)^2 \quad (05)$$

$$\Downarrow \quad \Downarrow$$

$$= (-) \quad (+) \quad / \quad (0)$$

$$\Delta \leq 0 \quad \text{තේ.} \quad \therefore \forall x, y \in \mathbb{R} \quad f(x, y) \geq 0 \quad \text{තේ.} \quad (05)$$

නිතරනය $y = -1$ තිව ය. (05)

$$0 = x^2 - 2x + 3 - 6x + 2 + 11$$

$$0 = x^2 - 8x + 16 = (x-4)^2 \quad (05)$$

$$x = 4 \quad \text{තේ} \quad (05)$$

$$x = 4 \quad \text{නි} \quad y = -1 \quad \text{තිව} \quad f(x, y) = 0 \quad \text{තේ.}$$

$$c) f(x) = \phi(x)(x-1)(x-2)(x-3) + \lambda(x-2)(x-3) + \mu(x-1)(x-3) + \tau(x-1)(x-2) \quad (05)$$

$$x = 1 \text{ ସଂଖ୍ୟା};$$

$$f(1) = \lambda(-1)(-2)$$

$$\lambda = \frac{f(1)}{2} \quad (05)$$

$$x = 2 \text{ ସଂଖ୍ୟା};$$

$$f(2) = \mu(1)(-1)$$

$$\mu = \frac{f(2)}{-1} = -f(2) \quad (05)$$

$$x = 3 \text{ ସଂଖ୍ୟା};$$

$$f(3) = \tau(2)(1)$$

$$\tau = \frac{f(3)}{2} \quad (05)$$

$$f(x) = (x^5 + kx^2)$$

$$\lambda = \frac{1}{2} f(1) = \frac{1}{2} (1+k) \quad (05)$$

$$\mu = -f(2) = -(32+4k) \quad (05)$$

$$\tau = \frac{f(3)}{2} = \frac{1}{2} (243+9k) \quad (05)$$

$$\therefore \text{ ଗଠନ } = \lambda(x-2)(x-3) + \mu(x-1)(x-3) + \tau(x-1)(x-2)$$

$$x^0; = 0$$

$$\lambda + \mu + \tau = 0 \quad (05)$$

$$\frac{1}{2} (1+k) + (-32-4k) + \frac{1}{2} (243+9k) = 0 \quad (05)$$

$$k = -90 \quad (05)$$

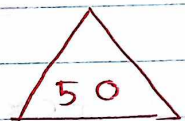
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12

a) අඩුතම දිශ්ඡතාව දිශ්චන ස්වභාවය } 6C_2 (10)
 තෝරා ගත් 2 තෝරා ගත හැක්
 වෙනස් තත්ත්වයන් ගණන } = 15 (10)

අඩුතම දිශ්ඡතාව දිශ්චන ස්වභාවය } 7C_4 (10)
 තෝරා නොගෙන ගත හැක්
 තත්ත්වයන් ගණන } = 35 (10)

∴ මෙලෙස තෝරා ගත හැක්
 වෙනස් ඡන්ද තත්ත්වයන්
 ගණන } $15 + 35$
 = 50 (10)



$$b) \quad U_r = \frac{2r-1}{r(r+1)(r+2)}$$

$$r U_r = \frac{2r-1}{(r+1)(r+2)} \rightarrow \textcircled{1} \quad (05)$$

$$U_{r+1} = \frac{2r+1}{(r+1)(r+2)(r+3)} \quad (05)$$

$$(r+3) U_{r+1} = \frac{2r+1}{(r+1)(r+2)} \rightarrow \textcircled{2}$$

$$\textcircled{2} - \textcircled{1}$$

$$\begin{aligned} (r+3) U_{r+1} - r U_r &= \frac{2r+1}{(r+1)(r+2)} - \frac{2r-1}{(r+1)(r+2)} \\ &= \frac{2}{(r+1)(r+2)} \quad (05) \end{aligned}$$

$$\therefore \underline{\underline{k = 2}} \quad (05)$$

$$(r+1) U_{r+1} - r U_r + 2 U_{r+1} = 2 \left[\frac{1}{r+1} - \frac{1}{r+2} \right] \quad (05)$$

$$\therefore 2 U_{r+1} = 2 \left[\frac{1}{r+1} - \frac{1}{r+2} \right] + \left[r U_r - (r+1) U_{r+1} \right] \quad (05)$$

$$\therefore r=1 ; 2u_2 = 2\left(\frac{1}{2} - \frac{1}{3}\right) - [u_1 - 2u_2]$$

$$r=2 ; 2u_3 = 2\left(\frac{1}{3} - \frac{1}{4}\right) - [2u_2 - 3u_3]$$

$$r=3 ; 2u_4 = 2\left(\frac{1}{4} - \frac{1}{5}\right) - [3u_3 - 4u_4]$$

(05)

$$r=n-2 ; 2u_{n-1} = 2\left(\frac{1}{n-1} - \frac{1}{n}\right) - [(n-2)u_{n-2} - (n-1)u_{n-1}]$$

$$r=n-1 ; 2u_n = 2\left(\frac{1}{n} - \frac{1}{n+1}\right) - [(n-1)u_{n-1} - nu_n]$$

(05)

$$2 \sum_{r=1}^n u_r = 2u_1 + 2\left(\frac{1}{2} - \frac{1}{n+1}\right) + [u_1 - nu_n]$$

(05)

$$\sum_{r=1}^n u_r = u_1 + \frac{1}{2} - \frac{1}{n+1} + \frac{1}{2} [u_1 - nu_n]$$

$$\sum_{r=1}^n u_r = \frac{3 \times u_1}{2} + \frac{1}{2} - \left(\frac{nu_n}{2} + \frac{1}{n+1} \right)$$

(05)

(05)

$$\sum_{r=1}^n u_r = \frac{3 \times 1}{2 \times 2 \times 3} + \frac{1}{2} - \left(\frac{nu_n(n+1)}{2(n+1)} + \frac{2}{2(n+1)} \right)$$

(05)

$$= \frac{3}{4} - \frac{n(n+1)}{2(n+1)} \left\{ \frac{(2n-1)}{n(n+1)(n+2)} \right\} - \frac{2}{2(n+1)}$$

$$= \frac{3}{4} - \frac{(2n-1)}{2(n+1)(n+2)} - \frac{2}{2(n+1)}$$

$$= \frac{3}{4} - \left[\frac{(2n-1) + 2(n+2)}{2(n+1)(n+2)} \right] \quad (05)$$

$$\sum_{r=1}^n u_r = \frac{3}{4} - \frac{(4n+3)}{2(n+1)(n+2)} \quad (05)$$

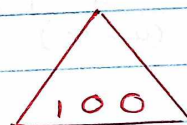
$$\lim_{n \rightarrow \infty} \sum_{r=1}^n u_r = \frac{3}{4} - \frac{1}{2} \lim_{n \rightarrow \infty} \frac{(4/n + 3/n^2)}{(1 + 3/n + 2/n^2)} \quad (05)$$

$$= \frac{3}{4} - \frac{1}{2} (0) \quad (05)$$

$$= \frac{3}{4}$$

$\therefore \sum_{r=1}^{\infty} u_r$ శరణు, 0 అవుతుంది. (05)

$$\sum_{r=1}^{\infty} u_r = \frac{3}{4} \quad (05)$$



13

$$a) A = \begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix}$$

$$P = \begin{pmatrix} 1 & 1 \\ x & y \end{pmatrix}$$

$$P^{-1} = \frac{1}{(y-x)} \begin{pmatrix} y & -1 \\ -x & 1 \end{pmatrix}$$

$$AP = \begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ x & y \end{pmatrix}$$

$$AP = \begin{pmatrix} -1+2x & -1+2y \\ 4+x & 4+y \end{pmatrix}$$

$$P^{-1}AP = \begin{pmatrix} \frac{y}{y-x} & \frac{-1}{y-x} \\ \frac{-x}{y-x} & \frac{1}{y-x} \end{pmatrix} \begin{pmatrix} -1+2x & -1+2y \\ 4+x & 4+y \end{pmatrix}$$

විකේන්ද්‍රක කොන්සාන්ස් කොන්සා

$$* \frac{y}{(y-x)} (2y-1) + \frac{(-1)}{(y-x)} (4+y) = 0$$

$$y-x \neq 0$$

$$2y^2 - y - 4 - y = 0$$

$$y^2 - y - 2 = 0$$

$$(y-2)(y+1) = 0$$

$$y = 2 \quad \text{හෝ} \quad y = -1$$

$$* \frac{-x}{y-x} (-1+2x) + \frac{1}{(y-x)} (4+x) = 0$$

$$x - 2x^2 + 4 + x = 0$$

$$x^2 - x - 2 = 0$$

$$x = 2 \quad \text{or} \quad x = -1$$

$$\therefore x < y$$

$$\underline{\underline{x = -1 \quad y = 2}}$$

A 4

$\Rightarrow$

$$A^2 = \begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1+8 & -2+2 \\ -4+4 & 8+1 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix}$$

$$A^3 = \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -9 & 18 \\ 36 & 9 \end{pmatrix}$$

$$A^4 = 9 \begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 4 & 1 \end{pmatrix}$$

$$= 9 \begin{pmatrix} 1+8 & -2+2 \\ -4+4 & 8+1 \end{pmatrix} \Rightarrow$$

$$A^4 = 9 \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix}$$

$$A^4 = 81 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$b) \left| \frac{z_1 - z_2}{z_1 + z_2} \right| = 1$$

$$\frac{|z_1 - z_2|}{|z_1 + z_2|} = 1$$

$$|z_1 - z_2| = |z_1 + z_2|$$

$$\left| z_2 \left[\left(\frac{z_1}{z_2} \right) - 1 \right] \right| = \left| z_2 \left[\left(\frac{z_1}{z_2} \right) + 1 \right] \right|$$

$$\left| \frac{z_1}{z_2} - 1 \right| = \left| \frac{z_1}{z_2} + 1 \right| = \left| \frac{z_1}{z_2} - (-1) \right|$$

$\therefore \frac{z_1}{z_2}$ යනු ඍණාත්මක සංඛ්‍යාවකි.

$$A = (0, 1) \quad B = (0, -1) \quad \text{විට}$$

$\frac{z_1}{z_2}$ A B න් අර්ධ භ්‍රමණයක් මත පවතී.

$$\therefore \left| \text{Arg } \frac{z_1}{z_2} \right| = \frac{\pi}{2}$$

14

$$a) f(x) = \frac{ax + b}{(x+2)^2}$$

x විෂයේ අවකලනය

$$f'(x) = \frac{(x+2)^2(a) - (ax+b)2(x+2)}{(x+2)^4}$$

$$= \frac{a(x+2) - 2(ax+b)}{(x+2)^3}$$

$$f'(x) = \frac{2a - 2b - ax}{(x+2)^3} \quad (10)$$

$$(0, 1/4) \text{ දී } f'(x) = 0 \text{ වේ.}$$

$$2a - 2b - ax = 0$$

$$2a - 2b = 0$$

$$a - b = 0 \rightarrow \textcircled{1} \quad (05)$$

$$(0, 1/4) \text{ } f(x) \text{ ට අවමය}$$

$$1/4 = \frac{b}{(2)^2} \quad (05)$$

$$\underline{\underline{b = 1}}$$

$$\underline{\underline{a = 1}} \quad (05)$$

පරික්ෂා කිරීමේදී ඔබගේ නම සඳහන් කරන්න

$$x + 2 = 0$$

$$\underline{x = -2}$$

(05)

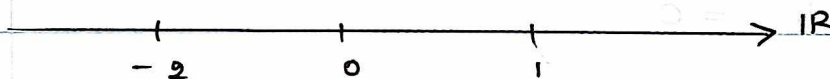
හරයේ ලක්ෂ්‍යය = $(0, 1/4)$

නැවතුම් ලක්ෂ්‍යය සඳහා $f''(x) = 0$

$$\frac{2(x-1)}{(x+2)^4} = 0$$

$$\underline{x = 1}$$

$$f(x) = \frac{2}{(3)^2} = \frac{2}{9} \quad (1, 2/9) \quad (05)$$



(15)

	$x < -2$	$-2 < x < 0$	$0 < x < 1$	$x > 1$
$f'(x)$ $= \frac{-x}{(x+2)^3}$	(-) අඩු වේ	(+) වැඩි වේ	(-) අඩු වේ	(-) අඩු වේ.
$f''(x)$ $= \frac{2(x-1)}{(x+2)^4}$	(-) යථ අවතල	(-) යථ අවතල	(-) යථ අවතල	(+) උඩු අවතල

(+) (-) $\rightarrow$ IR $x = 0$ දී උපරිම ලක්ෂ්‍යයකි

$x = 1$ දී අවමයකි. මාරු වන ත්‍යාගය නැවතුම් ලක්ෂ්‍යයකි.

$x = 1$ දී $f'(x)$ තීරණය

$$f'(x) = \frac{-1}{(1+2)^3} < 0 \text{ වේ. } (05)$$

$x \rightarrow \pm \infty$ කාණ්ඩය (05)

$$f(x) = \frac{x+1}{x^2+4x+4} = \frac{1/x + 1/x^2}{1 + 4/x + 4/x^2}$$

$x \rightarrow +\infty$

$$f(x) = \frac{1/x + 1/x^2}{1 + 4/x + 4/x^2} = 0 \quad \begin{array}{l} x \rightarrow +\infty \\ \underline{\underline{f(x) \rightarrow 0}} \end{array}$$

$x \rightarrow -\infty$

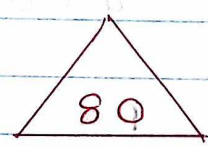
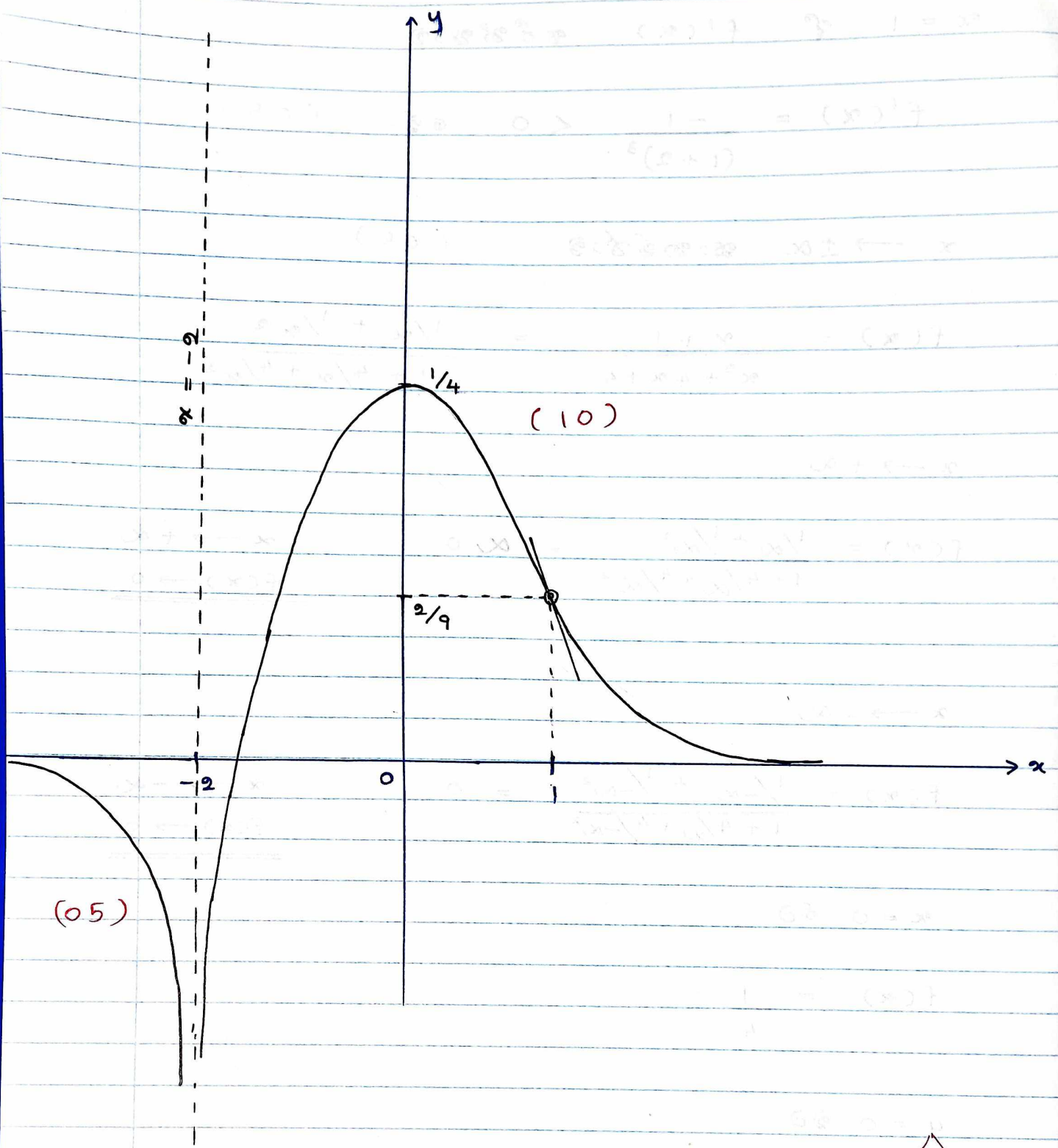
$$f(x) = \frac{1/-\infty + 1/-\infty^2}{1 + 4/-\infty + 4/-\infty^2} = 0 \quad \begin{array}{l} x \rightarrow -\infty \\ \underline{\underline{f(x) \rightarrow 0}} \end{array}$$

$x = 0$ වේ

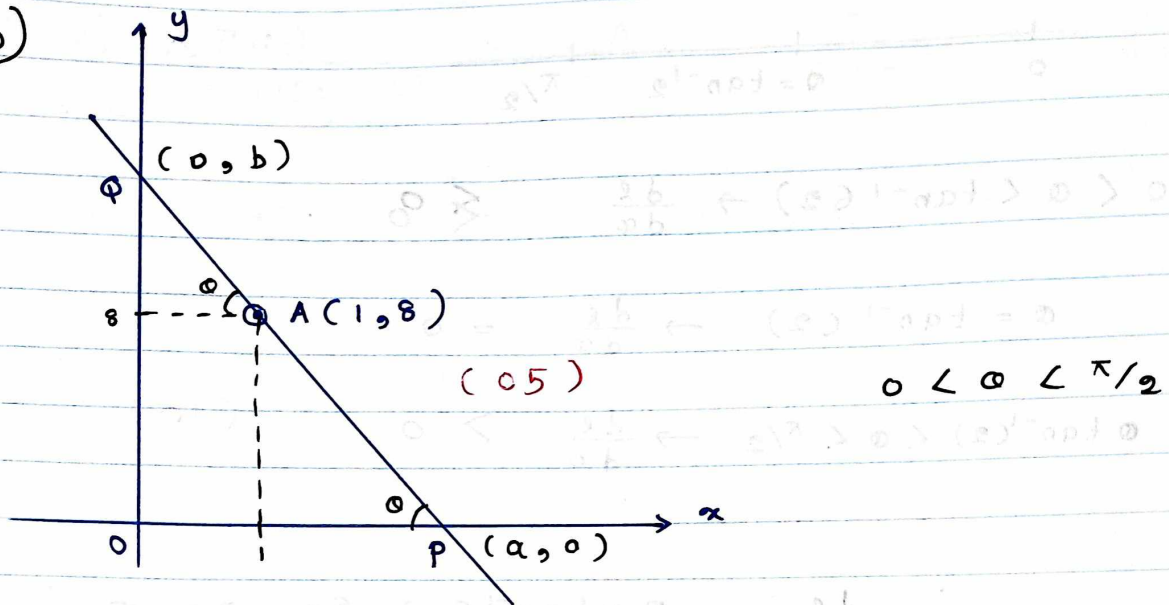
$$f(x) = \frac{1}{4}$$

$y = 0$ වේ

$$x = -1 \quad (0, -1) \quad (05)$$



b)



$$PQ = PA + AQ \quad (05)$$

$$l = 8 \operatorname{cosec} \theta + \sec \theta \quad (05)$$

0 විෂයයන් අවමකරණය

$$\frac{dl}{d\theta} = -8 \operatorname{cosec} \theta \cot \theta + \sec \theta \tan \theta \quad (10)$$

$$\frac{dl}{d\theta} = \frac{-8 \cos \theta}{\sin^2 \theta} + \frac{\sin \theta}{\cos^2 \theta} \quad (05)$$

$$\frac{dl}{d\theta} = 0 \quad \text{වි} \quad (05)$$

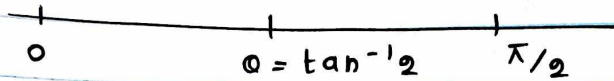
$$\frac{\sin \theta}{\cos^2 \theta} - \frac{8 \cos \theta}{\sin^2 \theta} = 0$$

$$\sin^3 \theta = 8 \cos^3 \theta$$

$$\tan^3 \theta = 8$$

$$\tan \theta = 2 \quad (05) \quad [0 < \theta < \pi/2]$$

$$\theta = \tan^{-1}(2)$$



(05)

$$0 < \theta < \tan^{-1}(2) \rightarrow \frac{dL}{d\theta} < 0$$

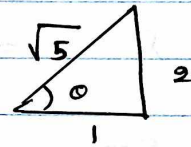
$$\theta = \tan^{-1}(2) \rightarrow \frac{dL}{d\theta} = 0$$

$$\tan^{-1}(2) < \theta < \pi/2 \rightarrow \frac{dL}{d\theta} > 0$$

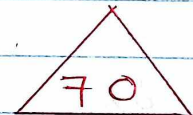
(10)

$\therefore \theta = \tan^{-1}(2)$ නම් PQ ඉඟ
අඟම වේ. (05)

$$PQ \text{ අඟම} = 8 \times \frac{\sqrt{5}}{2} + \sqrt{5}$$



$$= \underline{\underline{5\sqrt{5} \text{ මීටර්}}} \quad (05)$$



15

$$a) \frac{4}{(x-1)(x+1)^2} = \frac{A}{(x-1)} + \frac{B}{(x+1)} + \frac{C}{(x+1)^2} \quad (05)$$

$$4 = A(x+1)^2 + B(x+1)(x-1) + C(x-1) \quad (05)$$

$$x^2; \quad A + B = 0 \rightarrow (1)$$

$$x; \quad 2A + C = 0 \rightarrow (2)$$

$$x^0; \quad A - B - C = 4 \rightarrow (3)$$

$$(1) + (3) \quad 2A - C = 4 \rightarrow (4)$$

$$(2) + (4) \quad 4A = 4$$

$$\underline{\underline{A = 1}}$$

$$\underline{\underline{B = -1}} \quad (10)$$

$$\underline{\underline{C = -2}}$$

$$\therefore \frac{4}{(x-1)(x+1)^2} = \frac{1}{(x-1)} - \frac{1}{(x+1)} - \frac{2}{(x+1)^2} \quad (05)$$

$$I = \int \frac{1}{(1-e^x)(1+e^x)^2} dx$$

$$t = e^x \quad \text{అలా అక్షరం}$$

$$\frac{dt}{dx} = e^x$$

$$1 = e^x \frac{dx}{dt} = t \frac{dx}{dt}$$

$$\therefore I = \int \frac{1}{(1-1/t)(1+t)^2} dx$$

$$I = \int \frac{1}{(t-1)(t+1)^2} \cdot t dx$$

$$I = \int \frac{1}{(t-1)(t+1)^2} dt$$

$$I = \frac{1}{4} \int \frac{4}{(t-1)(t+1)^2} dt$$

$$I = \frac{1}{4} \left\{ \int \frac{1}{(t-1)} dt - \int \frac{1}{(t+1)} dt - 2 \int \frac{1}{(t+1)^2} dt \right\}$$

$$I = \frac{1}{4} \left\{ \ln |t-1| - \ln |t+1| + 2 \frac{1}{(t+1)} \right\}$$

$$I = \frac{1}{4} \ln \frac{|t-1|}{|t+1|} + \frac{1}{2} \frac{1}{(t+1)}$$

$$I = \frac{1}{4} \ln \frac{|e^x-1|}{|e^x+1|} + \frac{1}{2} \frac{1}{(e^x+1)} + c$$

$$b) \quad I_1 = \int \frac{x - \sin x}{1 - \cos x} dx$$

$$= \int \frac{x - \sin x}{2 \sin^2 x/2} dx$$

$$I = \frac{1}{2} \left[\int \frac{x dx}{\sin^2 x/2} - \int \frac{2 \cos^{x/2}}{\sin^{x/2}} dx \right]$$

$$I = \frac{1}{2} \left[\int \frac{x dx}{\sin^2 x/2} - \int \cot^{x/2} dx \right]$$

$$I = \frac{1}{2} \left[\underbrace{\int x \operatorname{cosec}^2 x/2 dx}_{I_2} - \int \cot^{x/2} dx \right]$$

$$I_2 = \int x \operatorname{cosec}^2 x/2 dx$$

$$= 2 \int x \frac{d(-\cot^{1/2} x)}{dx} dx$$

$$= 2 \left\{ -x \cot^{1/2} x + \int \cot^{1/2} x dx \right\}$$

$$I_2 = 2 \left\{ -x \cot^{1/2} x + 2 \ln |\sin^{1/2} x| \right\}$$

$$\therefore I = \frac{1}{2} \times 2 \left\{ -x \cot^{1/2} x + 2 \ln |\sin^{1/2} x| \right\} - \int \cot^{x/2} dx$$

$$I = -x \cot^{x/2} + 2 \ln |\sin^{x/2}| - 2 \ln |\sin^{x/2}|$$

$$I = -x \cot x/2 + C$$

(c යනු අනිකුත්
නියතයන්)

$$\left[\begin{array}{c} x^b e^{1/x} \log x \\ e^{1/x} \log x \end{array} \right] \cdot \left[\begin{array}{c} -x^b x \\ e^{1/x} \log x \end{array} \right] \cdot \frac{1}{x} = I$$

$$\left[\begin{array}{c} x^b e^{1/x} \log x \\ e^{1/x} \log x \end{array} \right] \cdot \left[\begin{array}{c} -x^b x \\ e^{1/x} \log x \end{array} \right] \cdot \frac{1}{x} = I$$

$$\left[\begin{array}{c} x^b e^{1/x} \log x \\ e^{1/x} \log x \end{array} \right] \cdot \left(\begin{array}{c} x^b e^{1/x} \log x \\ e^{1/x} \log x \end{array} \right) \cdot \frac{1}{x} = I$$

$$\left[\begin{array}{c} x^b e^{1/x} \log x \\ e^{1/x} \log x \end{array} \right] \cdot \frac{1}{x} = I$$

$$\left[\begin{array}{c} x^b (x e^{1/x} \log x) \\ x^b \end{array} \right] \cdot \frac{1}{x} =$$

$$\left\{ \begin{array}{c} x^b x e^{1/x} \log x \\ x^b \end{array} \right\} \cdot \frac{1}{x} =$$

$$\left\{ \begin{array}{c} |x e^{1/x} \log x| \log x + x e^{1/x} \log x \\ x^b \end{array} \right\} \cdot \frac{1}{x} = I$$

$$\left[\begin{array}{c} x^b e^{1/x} \log x \\ x^b \end{array} \right] \cdot \left\{ \begin{array}{c} |x e^{1/x} \log x| \log x + x e^{1/x} \log x \\ x^b \end{array} \right\} \cdot \frac{1}{x} = I$$

$$|x e^{1/x} \log x| \log x - |x e^{1/x} \log x| \log x + x e^{1/x} \log x = I$$

c) $\int_0^a f(a-x) dx$ $(a-x) = t$ ආදේශය ගනමු.

$x = 0$ විට $t = a - x = a - 0 = \underline{a}$

$x = a$ විට $t = a - x = a - a = 0$

$t = a - x$ නිසා x විෂයෙන් අවකලනය

$\frac{dt}{dx} = -1$

$-dt = dx$

R.H.S. = $\int_0^a f(a-x) dx$

= $\int_a^0 f(t) [-dt]$

= $-\int_a^0 f(t) dt$

= $\int_0^a f(t) dt$

= $\int_0^a f(x) dx$

= L.H.S.

$\therefore \int_0^a f(x) dx = \int_0^a f(a-x) dx$

$$I = \int_0^{\pi/2} \frac{x}{(\sin x + \cos x)^2} dx$$

$$I = \int_0^{\pi/2} \frac{(\pi/2 - x)}{(\cos x + \sin x)^2} dx \quad (05)$$

$$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{(\sin x + \cos x)^2} dx \quad (05)$$

$$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{1 + 2 \sin x \cos x} dx$$

$$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{2 \cos^2(\pi/4 - x)} dx \quad (05)$$

$$\sin x + \cos x = \sqrt{2} \cos(\pi/4 - x) \quad (05)$$

$$2I = \frac{\pi}{2} \int_0^{\pi/2} \frac{1}{2} \sec^2(x - \pi/4) dx \quad (05)$$

$$2I = \frac{\pi}{2} \left[\frac{1}{2} \tan(x - \pi/4) \right]_0^{\pi/2} \quad (05)$$

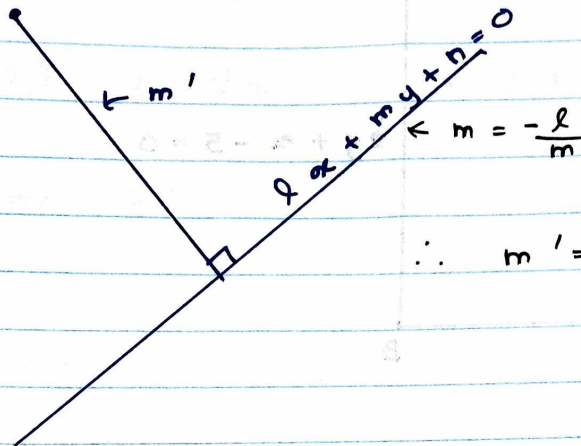
$$I = \frac{\pi}{8} \left\{ \tan \pi/4 - \tan(-\pi/4) \right\} \quad (05)$$

$$I = \frac{\pi}{8} \left\{ 2 \tan \pi/4 \right\}$$

$$\underline{\underline{I = \frac{\pi}{4}}}$$

16

a)

 $P_0(x_0, y_0)$ 

$$\therefore m \times m' = -1$$

$$m = -\frac{l}{m}$$

$$\therefore m' = \frac{m}{l}$$

(2-1) A

උමුණක සේ නම් කරනාය

$$\frac{y - y_0}{x - x_0} = \frac{m}{l}$$

$$mx - ly + ly_0 - mx_0 = 0$$

තෝරා නමුත්තේදිකය එක ලකුණයක් $(\bar{x}, \bar{y})$ ලෙස ගනමු.

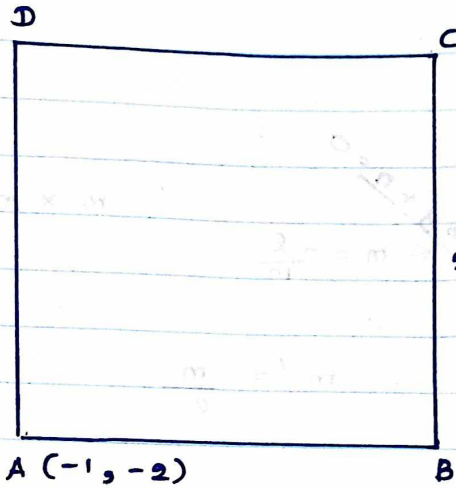
$$\frac{l\bar{x} + m\bar{y} + n}{\sqrt{l^2 + m^2}} = \pm \frac{(m\bar{x} - l\bar{y} + ly_0 - mx_0)}{\sqrt{m^2 + l^2}}$$

$\bar{x} \rightarrow x$ නා $\bar{y} \rightarrow y$ ලෙස ගෙන නිවැරදි කරමු

(+))

$$(l - m)x + (l + m)y + (n - ly_0 + mx_0) = 0$$

$$(l + m)x - (l - m)y + (n + ly_0 - mx_0) = 0$$



$$2y + x - 5 = 0$$

B D විකර්ණය යනු $\hat{A} \hat{B} C$ කෝණය මැනීමේදී
මේ.

මූලාශ්‍රය භාවිත කරමින්;

$$x_0 = -1, \quad y_0 = -2$$

$$l = 1, \quad m = 2, \quad n = -5$$

$$(1-2)x + (1+2)y + (-5+2-2) = 0$$

හෝ

$$(1+2)x + (2-1)y + (-5-2+2) = 0$$

$$\underline{-x + 3y - 5 = 0}, \quad \underline{3x + y - 5 = 0}$$

සමීකරණ 02 ක් ලැබේ. එනම් BD සරල
උණුදුම් 02 ක් ඇත.

No: _____

$$AD \text{ ସମୀକରଣ } = 2y + x + k = 0$$

$$(-1, -2) \text{ ବିନ୍ଦୁ } -4 - 1 + k = 0$$
$$k = 5$$

$$\underline{AD} \Rightarrow 2y + x + 5 = 0$$

D ଖୋଜିବା

$$2y + x + 5 = 0 \rightarrow \textcircled{1}$$

$$3x + y - 5 = 0 \rightarrow \textcircled{2}$$

$$\textcircled{2} \times 2 - \textcircled{1}$$

$$6x + 2y - 10 - 2y - x - 5 = 0$$

$$5x = 15$$

$$\underline{x = 3}$$

$$\textcircled{1} \text{ ଅ } 2y + 8 = 0$$

$$y = -4$$

$$\underline{\underline{D \equiv (3, -4)}}$$

$$2y + x + 5 = 0 \rightarrow \textcircled{1}$$

$$-x + 3y - 5 = 0 \rightarrow \textcircled{3}$$

$$\textcircled{1} + \textcircled{3} \quad 5y = 0$$

$$y = 0$$

$$\textcircled{3} \text{ ଅ } x = -5$$

$$\underline{\underline{D' \equiv (-5, 0)}}$$

A C ജി അളവുകൾ

$$y + 2 = \frac{1}{3}(x + 1)$$

$$(y + 2) = -3(x + 1)$$

$$x - 3y - 5 = 0$$

$$3x + y + 5 = 0$$

C കേൾ

$$x - 3y - 5 = 0 \rightarrow (4)$$

$$2y + x - 5 = 0 \rightarrow (5)$$

$$(5) - (4) \quad y = 0$$

$$x = 5$$

$$C \equiv (5, 0)$$

$$3x + y + 5 = 0 \rightarrow (6)$$

$$2y + x - 5 = 0 \rightarrow (5)$$

$$(5) \times 3 - (6)$$

$$6y + 3x - 15 - 3x - y - 5 = 0$$

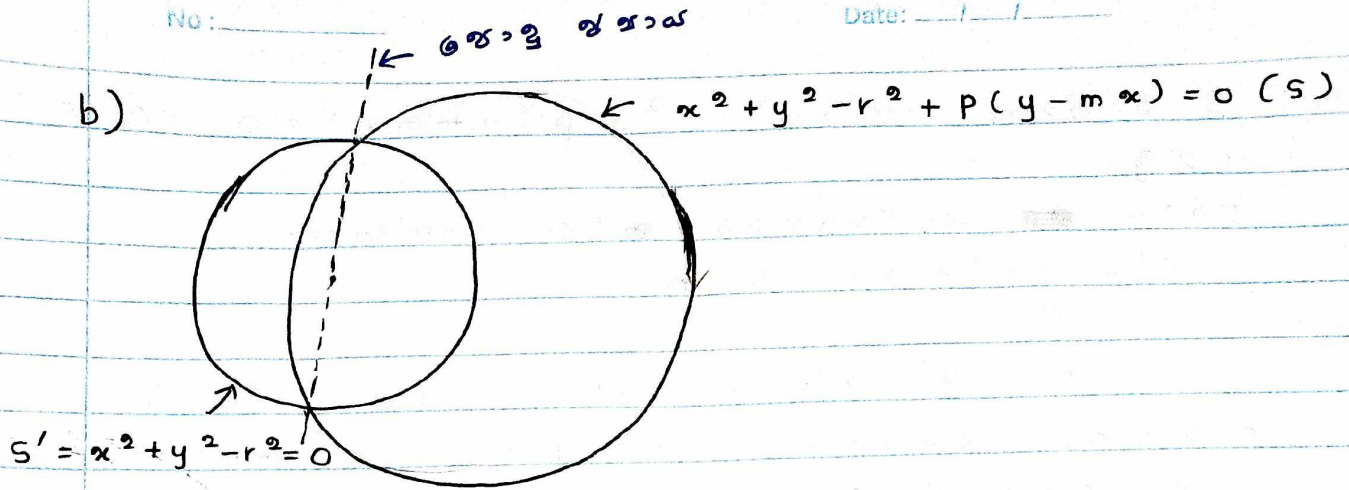
$$5y = 20$$

$$y = 4$$

$$x = -3$$

$$C' \equiv (-3, 4)$$

b)



S චක්‍රයේ මඟින් S' චක්‍රයේ ඉරික්කුණු -
 -කය කරයි. මේ ලෝමා ඉරාය S' චක්‍රයේ
 කේන්ද්‍රය කරන ගමන් කළ යුතුය. එය ඉතා
 ඉරික්කුණු කරයි.

* S' චක්‍රයේ කේන්ද්‍රය $O' \equiv (0, 0)$

ලෝමා ඉරායේ සමීකරණය

$$S - S' = 0$$

$$x^2 + y^2 - r^2 + p(y - mx) - (x^2 + y^2 - r'^2) = 0$$

$$p(y - mx) = 0$$

$$O' \equiv (0, 0) \text{ ආදේශය.}$$

$$L.H.S. = p(0 - m(0)) = 0 = R.H.S. \text{ වේ.}$$

එනම් ලෝමා ඉරාය $(0, 0)$ කරන යන නිසා
 $S' = 0$ චක්‍රයේ ඉරික්කුණු කළ යුතුය.

(ii) S චක්‍රය $x^2 + y^2 - r^2 + p(y - mx) = 0$ ලෙස ගනමු.

ඉරිඳි සමන්විත චක්‍රය

$$S' = 3x^2 + 3y^2 - 5 = 0$$

↓

$$x^2 + y^2 - 5/3 = 0 \text{ ලෙස ගනමු. එවිට } S$$

චක්‍රයේ සමීකරණය $x^2 + y^2 - \frac{5}{3} + p(y - mx) = 0$ ලෙස ලිවිය හැකිය.

$$x^2 + y^2 - pmx + py - 5/3 = 0$$

මෙහි කේන්ද්‍රය $= (-g, -f)$ නම්

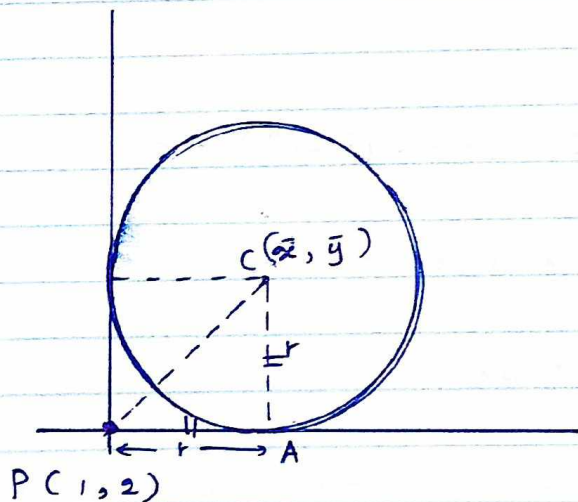
$$2g = -pm$$

$$2f = p$$

$$g = -\frac{pm}{2}$$

$$f = \frac{p}{2}$$

$$\therefore \text{කේන්ද්‍රය} = \left(\frac{pm}{2}, -\frac{p}{2} \right) = (\bar{x}, \bar{y})$$



$$S = x^2 + y^2 - pmx + py - \frac{5}{3} = 0$$

$$g = -\frac{pm}{2}, \quad f = \frac{p}{2}, \quad c = -\frac{5}{3}$$

$$r = \sqrt{g^2 + f^2 - c}$$

$$r = \sqrt{\left(-\frac{pm}{2}\right)^2 + \left(\frac{p}{2}\right)^2 - (-5/3)}$$

$$r = \sqrt{(-\bar{x})^2 + (-\bar{y})^2 - (-5/3)}$$

$$pc = r^2 + r^2$$

$$(\bar{x} - 1)^2 + (\bar{y} - 2)^2 = 2(\bar{x}^2 + \bar{y}^2 + 5/3)$$

$$\bar{x}^2 + \bar{y}^2 - 2\bar{x} + 1 - 4\bar{y} + 4 = 2\bar{x}^2 + 2\bar{y}^2 + \frac{10}{3}$$

$$\bar{x}^2 + \bar{y}^2 + 2\bar{x} + 4\bar{y} - \frac{5}{3} = 0$$

$$\therefore \text{କେନ୍ଦ୍ରର ସମୀକରଣ} \Rightarrow \underline{\underline{x^2 + y^2 + 2x + 4y - \frac{5}{3}}}$$

$$\underline{\underline{\text{କେନ୍ଦ୍ର} = (-1, -2)}}$$

$$\text{ଆର = } \sqrt{1 + 4 - (-5/3)}$$

$$= \sqrt{5 + 5/3}$$

$$= \underline{\underline{\sqrt{\frac{20}{3}}}}$$

17

$$a) \quad 4 - 4(\cos x - \sin x) - \sin 2x = 0$$

$$t = \cos x - \sin x$$

$$t^2 = 1 - \sin 2x$$

$$\therefore 4 - 4t - (1 - t^2) = 0$$

$$t^2 - 4t + 3 = 0$$

$$(t - 3)(t - 1) = 0$$

$$t = 3$$

or

$$t = 1$$

$$\cos x - \sin x = 3$$

$$\cos x - \sin x = 1$$

$$\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x = \frac{3}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} \cos x - \frac{1}{\sqrt{2}} \sin x = \frac{1}{\sqrt{2}}$$

$$\cos(x + \pi/4) = \frac{3}{\sqrt{2}} > 1$$

$$\cos(x + \pi/4) = \cos \pi/4$$

$$\underline{t \neq 3}$$

$$x + \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$$

$$x = 2n\pi \pm \frac{\pi}{4} - \frac{\pi}{4}$$

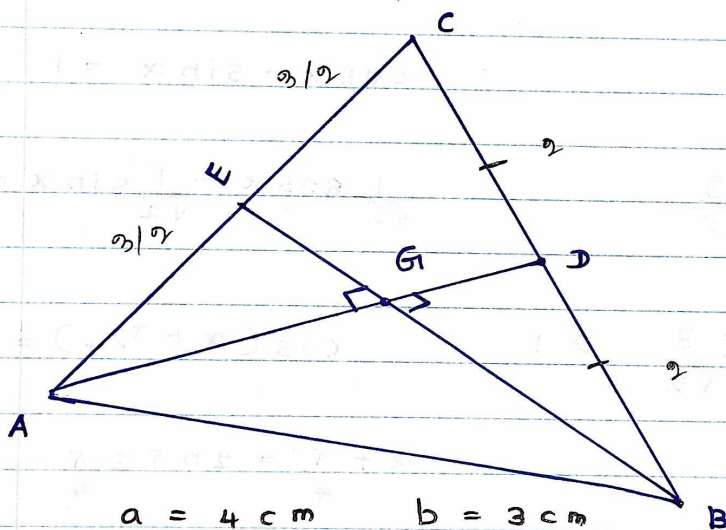
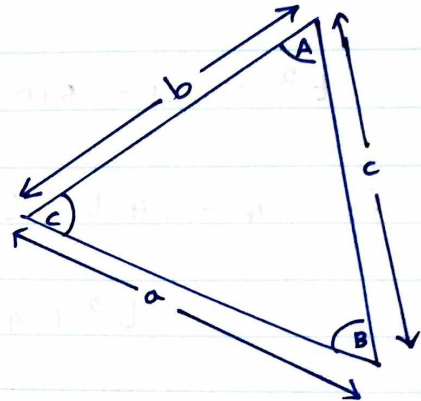
$$(n \in \mathbb{Z})$$

b) ඡුරුරු අංකනෙයන් මතම $ABC \Delta$ ඡු
ඡඳුනා...

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



$$\angle BGD = \pi/2$$

$$\angle EGD = \pi/2$$

$$ABC \Delta \Rightarrow c^2 = 4^2 + 3^2 - 2(4)(3)\cos C$$

$$c^2 = 25 - 24\cos C \rightarrow (1)$$

$$ADC \Delta \Rightarrow AD^2 = 3^2 + 2^2 - 2(2)(3)\cos C$$

$$AD^2 = 13 - 12\cos C \rightarrow (2)$$

$$BEC \triangle \Rightarrow BE^2 = 4^2 + \left(\frac{3}{2}\right)^2 - 2(4)\left(\frac{3}{2}\right)\cos C$$

$$BE^2 = \frac{73}{4} - 12\cos C \rightarrow (3)$$

$$\therefore AD \perp BE$$

$$AB^2 = AG^2 + BG^2$$

$$c^2 = \left(\frac{2}{3}AD\right)^2 + \left(\frac{2}{3}BE\right)^2$$

$$c^2 = \frac{4}{9}(AD^2 + BE^2)$$

$$c^2 = \frac{4}{9} \left\{ 13 - 12\cos C + \frac{73}{4} - 12\cos C \right\}$$

$$c^2 = \frac{4}{9} \times \frac{1}{4} \left\{ 125 - 24\cos C \right\}$$

$$25 - 24\cos C = \frac{4}{36} \left(125 - 24\cos C \right)$$

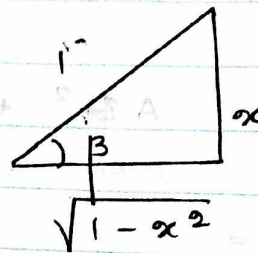
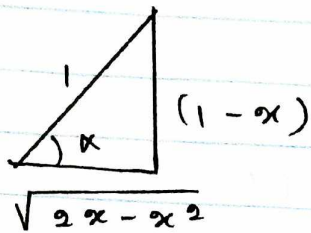
$$900 - 864\cos C = 500 - 96\cos C \times 4 \times 4$$

$$400 = 768\cos C$$

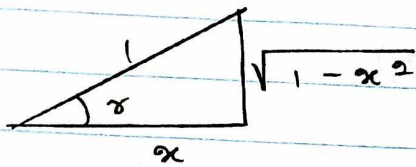
$$c) \underbrace{\sin^{-1}(1-x)}_{\alpha} + \underbrace{\sin^{-1}(x)}_{\beta} = \underbrace{\cos^{-1}(x)}_{\gamma}$$

$$\sin \alpha = (1-x)$$

$$\sin \beta = x$$



$$\cos \gamma = x$$



$$\alpha + \beta = \gamma$$

$$\cos(\alpha + \beta) = \cos \gamma$$

$$\cos \alpha \cos \beta - \sin \alpha \sin \beta = x$$

$$(\sqrt{2x-x^2})(\sqrt{1-x^2}) - (1-x)(x) = x$$

$$\sqrt{(2x-x^2)(1-x^2)} = x + (x-x^2)$$

$$(2x-x^2)(1-x^2) = [(2x-x^2)]^2$$

$$2x - 2x^3 - x^2 + x^4 = 4x^2 + x^4 - 4x^3$$

$$2x^3 - 5x^2 + 2x = 0$$

$$x = 0 \quad 2x^2 - 5x + 2 = 0$$

$$2x^2 - 5x + 2 = 0$$

$$2x^2 - 4x - x + 2 = 0$$

$$2x(x-2) - (x-2) = 0$$

$$2x - 1 = 0 \quad \text{അല്ല} \quad x - 2 = 0$$

$$x = 1/2 \quad \text{അല്ല} \quad x = 2 \quad (\text{ഒരു തൊണ്ട})$$

$$\therefore \underline{\underline{x = 0 \quad \text{അല്ല} \quad x = 1/2}}$$